COMPREHENSIVE OPERATIONS MANUAL

CodeCity AI

Complete Feature-by-Feature Operational Guide, Inputs, Outputs & Interactive Walkthrough

> *"See where your code will break next, and what breaks with it."*

This manual provides exhaustive instructions for exploring, operating, demonstrating, and evaluating every capability of CodeCity AI. Each section specifies exact user inputs, expected outputs, background mechanisms (ML + AST), and verification steps.

TARGET AUDIENCE

Software Engineers, Product Reviewers, Evaluators, Exhibition Visitors

RUNTIME ENVIRONMENT

FastAPI (Python 3.12+) • React 18 / Three.js • Port 5173 / 8000

OFFLINE DEMO REPOSITORIES

`pallets/flask`, `pallets/click`, `psf/requests`, `ecotrace-ai`

ML ENGINE & EXPLANATIONS

Zero-Leakage XGBoost • AST Blast Radius • Rule-Based SHAP

1 System Architecture & Quickstart

CodeCity AI transforms any Python Git repository into a real-time, interactive 3D cyber-metropolis. The application operates in a completely decoupled client-server architecture:

- **Backend (FastAPI & ML Engine)**: Traverses repository Git commit logs via PyDriller, parses Python Abstract Syntax Trees (`ast`) with NetworkX, computes cyclomatic complexity using Radon, and infers future bug probabilities via a pre-trained XGBoost model.
- **Frontend (React + Three.js)**: Renders hundreds of architectural buildings at 60 FPS using GPU instancing (`InstancedMesh`), procedural GLSL window facades, dynamic parabolic blast-radius tubes, and a cybernetic glassmorphism HUD.

One-Click Launch Command

CodeCity AI includes dedicated startup scripts for instantaneous local startup:

▶ LAUNCH INPUT

Double-click `start.bat` in the repository root, or run in terminal:

```
python run.py dev
```

✓ SYSTEM OUTPUT

- Backend activates at `http://127.0.0.1:8000`
- Frontend activates at `http://127.0.0.1:5173`
- Zero internet connection required

2 3D Cyber-City Navigation & Spatial Controls

The city occupies the entire viewport as the hero visual element. Code hierarchy maps directly into physical metropolitan urban structures:

Code Element	Metropolitan Visual Representation	Design Logic
Directories	City Districts & Glass Plates	Squarified treemap partition with floating 3D billboard labels.
Files (.py)	Architectural Skyscrapers	Chamfered corners, emissive rooftops, and procedural facade windows.
Lines of Code (LOC)	Building Footprint (Width & Depth)	Larger modules occupy larger city plots.
Cyclomatic Complexity	Building Height	Log-scaled vertical growth ($H \propto \ln(1 + ext{CC})$).
Defect Probability	Facade Emissive Color	Emerald Green (Safe) → Amber (Moderate) → Crimson (Critical).

Interactive Navigation Controls

USER GESTURES & INPUTS

- **Left-Click Drag**: Orbit camera around city center.
- **Right-Click Drag / Two-Finger Pan**: Pan camera across districts.
- **Mouse Wheel / Pinch**: Zoom in and out smoothly.
- **Hover Building**: Triggers cyan wireframe bracket and reveals tooltip.
- **Click Building**: Selects building, frames camera, triggers Blast Radius.
- `Esc`: Clears selection and returns to default angle.

VISUAL & SPATIAL OUTPUTS

- Silky 60 FPS camera fly-to animation.
- Dynamic wet asphalt ground reflections and cyber grid perspective.
- Selected building glows with electric cyan rooftop highlight.
- Unselected structures dim to 20% opacity to highlight focus.
- Slide-in Right Drawer opens with full file diagnostics.

3 Repository Switching & Offline Demo Datasets

CodeCity AI bundles four pre-mined demonstration repositories for instantaneous offline analysis. Users can explore dense production frameworks or lightweight scripts with zero waiting time.

How to Switch Repositories

INPUT METHODS

1. **Top Bar Dropdown**: Click the repository selector in the center-left of the top bar (ID: `nav-repo-select`).
2. **Command Palette**: Press `Ctrl+K`, type `flask`, `click`, `requests`, or `ecotrace`, and press `Enter`.

EXPECTED OUTPUT

- Instantaneous city layout generation (< 200ms).
- Minimap immediately reflects new building coordinates.
- Left Panel KPIs recalculate (Total Files, Avg Risk, Critical Hotspots).
- Hotspot list ranks new repository files from riskiest to safest.

Repository	Files	Architectural Characteristics	Recommended Exploration Goal
`pallets/flask`	10 files	Dense web framework core; high fan-in on `app.py` and `helpers.py`.	Examine cascading blast radius from `app.py` to dependent modules.
`pallets/click`	14 files	Command-line interface builder with deep parser and formatting modules.	Inspect cyclomatic complexity hotspots in `core.py` and `formatting.py`.
`psf/requests`	17 files	HTTP library with intricate session, adapter, and model dependencies.	Test 2-hop and 3-hop reverse reachability from `models.py`.
`ecotrace-ai`	4 files	Sparse repository demonstrating intelligent dynamic layout scaling ($2.4 imes$).	Observe how small codebases appear intentional and grand rather than empty.

4 Scanning Any Public GitHub or Local Repository

Beyond bundled datasets, CodeCity AI can mine and analyze any live Python repository from GitHub or your local disk.

Step-by-Step Scan Instructions

INPUT STEPS

1. In the Left Panel, locate the **Analyze Python Repository** card.
2. Paste any valid public Python Git URL (e.g., `https://github.com/psf/requests`) or enter a local folder path on your machine.
3. Click the blue **Scan** button or press `Enter`.

OUTPUT PIPELINE (ASYNCHRONOUS JOB)

- Submits POST request to `/api/analyze` → receives `job_id`.
- Status radar animation appears with progress bar:
  - *Cloning repository...* (0-20%)
  - *Mining commit history...* (20-60%)
  - *Parsing AST dependencies...* (60-85%)
  - *Executing XGBoost inference...* (85-100%)
- City immediately renders upon completion.

Safe Mining Guardrails

To ensure snappy local performance, the backend enforces a **150 MB repository size cap** and limits commit mining to the **latest 2,000 commits**.

5 Five Visualization Color Modes

CodeCity AI provides five distinct visual lenses to inspect different dimensions of code health. Each mode dynamically recalculates the building facade materials and updates the Bottom HUD legend.

Mode	Key	Data Source & Calculation	Color Spectrum & Tiers
1. Defect Risk	1	XGBoost ML predicted defect probability ($0.0 o 1.0$).	● Safe (<30%) • ● Moderate (30-60%) • ● Critical (>60%)
2. Complexity	2	Radon cyclomatic branching complexity (max CC score).	● Simple (≤4) • ● Moderate (5-10) • ● Complex (11-15) • ● Extreme (>15)
3. 90-Day Churn	3	Number of commit edits touching the file in prior 90 days.	● Low (≤5) • ● Moderate (6-20) • ● Hot (>20 edits)
4. File Age	4	Days since file was originally created in Git history.	● Fresh (≤30d) • ● Mature (>180d)
5. Impact Score	5	$ ext{Risk Probability} imes (1 + \ln(1 + ext{BlastRadius}))$.	● Low (<1.0) • ● Medium (1.0-3.0) • ● Severe (>3.0)

How to Switch Modes

INPUT METHODS

- **Keyboard Numbers**: Press `1`, `2`, `3`, `4`, or `5` on your keyboard.
- **Bottom HUD Pills**: Click any of the mode buttons in the center of the Bottom HUD.
- **Command Palette**: Press `Ctrl+K` → type "Switch to..." → press `Enter`.

OBSERVED OUTPUT

- Active pill illuminates with a glowing cyan outline.
- Bottom HUD legend dynamically renders exact score thresholds.
- All 3D buildings instantly transition their facade materials.
- Procedural window lights and roof trims adapt harmoniously.

6 Defect Hotspot Ranking & Bidirectional Sync

The Left Panel features a curated list of the top highest-risk files in the repository. It provides bidirectional synchronization with the 3D scene.

USER ACTIONS

- **Hover over any Hotspot item** in the Left Panel list.
- **Click any Hotspot item** in the Left Panel list.
- **Click any 3D Building** directly inside the city.

SYNCHRONIZED SYSTEM OUTPUT

- **Hovering**: Immediately displays a cyan scanning bracket and highlights the corresponding building in 3D.
- **Clicking**: Camera smoothly flies and frames the selected file; Right Drawer opens with detailed diagnostics; Blast Radius arcs illuminate.
- **Clicking Building in 3D**: Automatically scrolls the Left Panel hotspot list to highlight the selected file.

7 Blast Radius: The Signature Dependency Moment

The blast radius answers the core product question: **"What breaks when this file fails?"** Using an AST-derived directed dependency graph, CodeCity AI computes exact reverse-reachability up to 3 hops.

Visual Anatomy of the Blast Radius

TRIGGER INPUT

Click on any high-impact file (e.g. `app.py` in `pallets/flask` or `models.py` in `psf/requests`).

SPECTACULAR 3D OUTPUT

1. **Shockwave Ring**: An expanding cyan shockwave emits across the wet asphalt from the building's base.
2. **Parabolic Tubes**: Glowing 3D tubular arches rise and connect the selected module to dependent files.
3. **Glowing Data Packets**: Luminous particles stream along the tubes from origin to dependents.
4. **Hop-Depth Chromatic Coding**:
   - **Hop 1 (Direct Importers)**: Neon Cyan tubes.
   - **Hop 2 (Transitive Callers)**: Electric Violet tubes.
   - **Hop 3 (Distant Cascades)**: Cyber Rose tubes.
5. **Contextual Dimming**: Unrelated buildings dim to 20% opacity.

8 Slide-In Right Drawer & Explainable SHAP Diagnostics

When a building is selected, the Right Drawer slides in with zero-hallucination, deterministic model explanations.

Anatomy of File Diagnostics

Diagnostic Element	Description & Output Value
Animated Risk Gauge	Circular SVG gauge tweening from 0% to the exact predicted probability (e.g., **96% Defect Risk**).
Architectural Impact	Numerical impact score factoring risk and dependent count: $ ext{Risk} imes (1 + \ln(1 + ext{BlastRadius}))$.
AI Defect Diagnosis	Plain-English synthesis (e.g., *"High defect likelihood (96.2%). Driven primarily by large codebase footprint (17 LOC) and intricate branching complexity (max CC 5.0). Partially buffered by mature codebase longevity."*)
SHAP Feature Contributions	Granular breakdown of feature forces: • ▲ Positive Forces (Red): Increases defect risk (e.g. Lines of Code +1.58, High CC +1.51). • ▼ Mitigating Forces (Cyan): Decreases risk (e.g. File Age -0.22, Low Churn -0.53).
Dependent Files List	Interactive clickable list of direct and indirect callers affected by this file.

9 2D Radar Minimap & Command Palette

Real-Time 2D Radar Minimap (Bottom HUD)

INPUTS

- Look at the bottom-left corner of the Bottom HUD.
- Click the **Center** button on the minimap.

OUTPUTS

- Displays 2D scatter coordinates of all repository buildings.
- Active building pulses with a bright cyan beacon.
- Clicking "Center" smoothly resets camera angle to center framing.
- Displays active node count and selected filename.

Command Palette (`Ctrl+K` / `Cmd+K`)

HOW TO USE

1. Press `Ctrl+K` or click **Command** ⌘**K** in the top navbar.
2. Type any keyword (e.g., `risk`, `flask`, `eco`, `model`, `exhibition`).
3. Navigate with `↑` `↓` arrow keys and press `Enter`.
4. Press `Esc` to dismiss.

SUPPORTED COMMANDS

- Switch visualization modes (Risk, Complexity, Churn, Age, Impact).
- Switch repositories (Flask, Click, Requests, EcoTrace).
- Toggle Exhibition Mode & Low-Power Mode.
- Open Model Card evaluation modal.
- Reset 3D camera to default overview.

10 Model Evaluation Card (Honest Zero-Leakage Metrics)

Clicking **Model Card** in the top bar opens an exhaustive evaluation modal displaying genuine machine learning performance.

Evaluation Metrics & Zero Data Leakage

Metric	Value	Honest Interpretation
ROC-AUC	`0.667`	Solid predictive capability distinguishing bug-prone modules on strictly held-out repositories.
Precision @ 10	`0.500`	5 out of the top 10 flagged files experienced verified future defect commits.
Recall @ 10%	`0.450`	Captures nearly half of all future defects by inspecting only the top 10% riskiest files.
Benchmark vs Churn Baseline	`+0.12 ROC-AUC`	Demonstrably outperforms naive commit frequency ranking (Churn ROC-AUC: `0.547`).

> **Zero Data Leakage Protocol**
> Training enforced strict temporal cutoffs at time T. All historical features were extracted strictly before T. Defect labels were extracted from bug fixes in the future window $(T, T + 90 ext{ days}]$. Tested on a held-out repository (`httpie/cli`) never seen during training.

11 Exhibition Mode & Inactivity Attract Kiosk

Designed specifically for college project showcases, booth exhibitions, and conference demos, Exhibition Mode transforms CodeCity AI into a hands-free presentation kiosk.

How to Activate and Experience Exhibition Mode

ACTIVATION INPUTS

- Press `P` on your keyboard.
- Click **Exhibition** in the top navbar.
- **Auto-Attract**: Leave the keyboard and mouse untouched for **45 seconds**.

KIOSK PRESENTATION EXPERIENCE

- **Continuous Auto-Orbit**: Camera smoothly glides around the city at an elegant cinematic angle.
- **Rotating Hotspot Showcase**: Automatically cycles through the top 5 highest-risk files every 8 seconds.
- **Blast Radius Demonstration**: Conduits and flowing particles pulse automatically for each highlighted hotspot.
- **Projector HUD**: Displays high-contrast banners and risk callouts readable from 10+ feet away.
- **Instant Wakeup**: Any mouse motion, touch, or keypress immediately resumes normal interactive mode.

12 Eco / Low-Power Rendering Mode

Ensures smooth demonstration on older laptops, ultrabooks with integrated Intel/AMD graphics, and battery-powered devices.

HOW TO TOGGLE

Click the **Eco** button in the top navbar (ID: `nav-eco-toggle`) or press `Ctrl+K` → "Enable Low-Power Mode".

HARDWARE OPTIMIZATION RESULTS

- Bypasses multi-pass postprocessing bloom shaders.
- Lightens GPU fillrate load while preserving complete 3D geometry and color modes.
- Navbar and Bottom HUD badge indicate **Low-Power Active** in warm amber.
- Guarantees sustained 60 FPS even on low-end hardware.

13 Complete Master Shortcut Cheat Sheet

Key / Gesture	Feature Triggered	Operational Effect
`Left-Click Drag`	Camera Orbit	Rotates the 3D metropolis around the central urban core.
`Right-Click Drag`	Camera Pan	Pans across districts and neighboring modules.
`Mouse Wheel`	Camera Zoom	Smoothly moves closer to inspect building facades and rooftops.
`Click Building`	Select & Blast Radius	Focal camera fly-to, Right Drawer opening, shockwave & dependency tubes.
`Ctrl+K` / `Cmd+K`	Command Palette	Opens instant fuzzy-searchable launcher for repos, modes, and actions.
`Esc`	Deselect / Close	Closes drawers/modals, clears active building selection, resets camera.
`F`	Focus Search	Instantly jumps cursor into the Left Panel file search bar.
`1`	Risk Mode	Colors city buildings by XGBoost ML defect probability ($0.0 o 1.0$).
`2`	Complexity Mode	Colors city buildings by Radon cyclomatic branching complexity.
`3`	Churn Mode	Colors city buildings by 90-day Git commit edit frequency.
`4`	Age Mode	Colors city buildings by file creation age (Fresh vs Matured).
`5`	Impact Mode	Colors city buildings by Architectural Impact score.
`P`	Exhibition Mode	Toggles hands-free cinematic auto-orbit and rotating hotspot showcase.

14 3-Minute Live Demonstration Walkthrough Script

Follow this exact sequence when presenting CodeCity AI to evaluators, professors, or exhibition visitors:

1. **The Hook (0:00 - 0:30)**:
   Open `pallets/flask`. Show the full-screen city at night. Explain: "*This is not a mockup—it's an interactive 3D digital twin of a real Python codebase. Directories are districts, files are buildings, and height is cyclomatic complexity.*"
2. **The Prediction (0:30 - 1:15)**:
   Point to the top hotspot (`app.py`). Explain: "*Our XGBoost model predicts a 96% defect likelihood in the upcoming release cycle based on historical commit churn, complexity, and graph topology.*"
3. **The Signature Blast Radius (1:15 - 2:00)**:
   Click `app.py`. Watch the shockwave emit and 3D parabolic conduits arc to dependent modules. Show the glowing data packets flowing across the city. Highlight: "*It's not just risky—it has an Architectural Impact score of 2.02 because 2 critical modules break if app.py fails.*"
4. **Explainability & Zero Data Leakage (2:00 - 2:30)**:
   Inspect the Right Drawer SHAP bars. Show why it's risky (Lines of Code +1.58, High CC +1.51). Open the **Model Card** to prove ROC-AUC 0.667 on held-out data with zero leakage.
5. **Exhibition Kiosk Mode (2:30 - 3:00)**:
   Press `P` to trigger Exhibition Mode. Let the camera orbit smoothly while explaining: "*The platform even includes an automatic attract mode that self-demonstrates at exhibition booths.*"